

REMARKS ON SIX-TANGENTIAL CONICS TO SMOOTH PLANE SEXTICS

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ABSTRACT.

Let n be a positive integer, and let $N := \{1, \dots, n\}$. We identify each subset of N with the strictly increasing sequence of its elements. Let G be a subgroup of the symmetric group S_n . We define a total order $<_l$ on 2^N by declaring that $a <_l a'$ if and only if

$$|a| < |a'|, \quad \text{or} \quad (|a| = |a'| \text{ and } a <_{\text{lex}} a'),$$

where $<_{\text{lex}}$ denotes the lexicographic order. We say that $a \in 2^N$ is *G-minimal* if it is the least element of its orbit a^G . If a is *G-minimal*, then every prefix of a is also *G-minimal*.

Let A be a finite abelian group of size n . Fix a bijection $A \cong N$, and use it to equip 2^A with the total order defined above. Let G be a subgroup of $\text{Aut}(A)$, viewed as a subgroup of S_n via this bijection. Let P be a property of subgroups of A that is *G*-invariant and preserved under taking subgroups. A subgroup of A satisfying P is called a *P*-subgroup. Our aim is to list all *P*-subgroups of A modulo the action of G . For a *P*-subgroup B , let $\text{mg}(B)$ be the $<_l$ -minimal generating set of B . We define a total order $<_\beta$ on the set of all *P*-subgroups of A by

$$B <_\beta B' \iff \text{mg}(B) <_l \text{mg}(B').$$

A *P*-subgroup B is said to be *G*-minimal if B is $<_\beta$ -minimal in the G -orbit B^G of B by G . For every *G*-minimal *P*-subgroup $B \neq \{0\}$, there exists a strictly increasing sequence $[g_1, \dots, g_m]$ (where we use the order on A by the bijection $A \cong N$) such that

$$0 \subsetneq \langle g_1 \rangle \subsetneq \langle g_1, g_2 \rangle \subsetneq \dots \subsetneq \langle g_1, \dots, g_m \rangle = B$$

and that $\{g_1, \dots, g_m\}$ is *G*-minimal.

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